# Grounding Radiology Reports in 3D Chest CT: Results of the ReXGrounding Challenge

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## Abstract

Segmenting each reported finding in the image would allow findings to be measured and tracked over time and let AI systems generate reports whose findings are verifiable, but so far AI models have performed poorly at this task. To advance this task, we organized the ReXGrounding Challenge at MICCAI 2026, the first challenge dedicated to it. Given a non-contrast chest CT scan and a free-text finding, participants produced a 3D segmentation of that finding. The challenge used 2,992 training and 200 validation scans from the ReXGroundingCT dataset, and evaluated methods on a held-out test set of 300 scans with 582 findings, exhaustively annotated by board-certified radiologists. In total, 48 teams from 15 countries made 760 submissions, and the best method reached a Dice of 0.375 and a hit rate of 0.77, matching agreement between two independent radiologist groups (Dice 0.32, hit rate 0.76, on a 50-scan subset). On the 100 scans of the pre-challenge benchmark, the winner reached a Dice of 0.390, against 0.285 for VoxTell, the best model before the challenge, which five of the six best teams fine-tuned. Non-focal findings remained far harder to segment than focal ones, instance F1 stayed below inter-reader agreement (0.34 vs. 0.47), and models sometimes segmented the right abnormality in the wrong location. Top methods thus reach Dice values in the range of agreement between independent radiologist groups, but non-focal findings and adherence to the anatomical location described in the text remain open problems.

*Keywords:* Challenge, Chest CT, Visual grounding, Text-guided segmentation, Radiology reports, Vision–language models

## 1. Introduction

AI systems for medical image interpretation are advancing quickly, but their use in clinical practice depends on more than accuracy: their outputs must also be explainable and verifiable (Rajpurkar and Lungren, 2023; Silcox et al., 2024; Abgrall et al., 2024). Radiology reports fall short of this because they describe the location of each finding only in words, such as a "5 mm nodule in the posterior segment of the right upper lobe", without marking it on the scan, so a reader has to find it again before it can be measured, compared with prior studies or checked. Grounding each finding, that is, segmenting the region a sentence describes, would make reports, whether written

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by radiologists or generated by AI, verifiable against the image (Bannur et al., 2024; Baharoon et al., 2026a). The need is greatest in chest CT, which is performed in large volumes (about 20 million examinations a year in the United States alone (Smith-Bindman et al., 2025)) and in which a single scan often contains many co-occurring findings, from small nodules to diffuse parenchymal change.

Grounding a free-text finding in 3D differs from the fixed-class segmentation evaluated by most medical imaging benchmarks and challenges (Antonelli et al., 2022; Bilic et al., 2023; Heller et al., 2021; Ma et al., 2024a; Heller et al., 2023; Kavur et al., 2021; Dorent et al., 2024; Luo et al., 2025; Ma et al., 2024b): a model must read a sentence that specifies the target by size, laterality, lobe or segment and morphology, and decide which of several similar lesions it refers to. Existing datasets, models and challenges do not support this task. Large chest CT datasets pair scans with reports but lack voxel-level labels (Hamamci et al., 2026b), and grounding datasets either link sentences to anatomical regions rather than to findings (Zhang et al., 2025a), have fewer than 250 cases (Bai et al., 2024), or focus on 2D modalities such as chest X-ray with bounding boxes instead of segmentation annotations (Boecking et al., 2022; De Castro et al., 2025). Current text-promptable segmentation models, trained mostly on short category-level prompts, fail on the descriptive findings written in radiology reports (Zhao et al., 2025c,a,b; Du et al., 2024; Subramanian et al., 2026; Baharoon et al., 2025). Recent challenges that pair images with radiology reports evaluate report generation or the localization of fixed classes of abnormalities, rather than grounding the findings described in a report directly in the image (Xu et al., 2024; Hamamci et al., 2025, 2026a). We previously released ReXGroundingCT, the first large-scale dataset linking findings from radiology reports to 3D segmentations in chest CT (Baharoon et al., 2026a).

To address this gap, we organized the ReXGrounding Challenge (Figure 1) at MICCAI 2026 as the first challenge on 3D segmentation conditioned on free-text findings. Given a non-contrast chest CT and one finding from its report, participants segmented that finding in 3D. Methods were evaluated on 300 held-out scans with 582 findings, exhaustively annotated by board-certified radiologists, and 48 teams from 15 countries made 760 submissions. The challenge led to four main findings, which this paper reports following the BIAS guideline (Maier-Hein et al., 2020). First, the best method reached a Dice of 0.375 and a hit rate of 0.77, matching the agreement between two independent radiologist groups. On the 100 scans of the pre-challenge benchmark it reached a Dice of 0.390, up from 0.285 for the best model before the challenge (Rokuss et al., 2026). Second, the top four teams were within 0.015 Dice and could not be separated statistically, so the choice of ranking scheme determined the winner. Third, the top methods converged on a common recipe: fine-tuning a pretrained text-promptable model, large anatomical context, text augmentation, ensembling, and post-processing to restrict masks to the lobe or side named in the sentence. Fourth, the overall Dice hides clear remaining gaps (Figure 3), in non-focal findings, in finding every instance of a multi-instance finding, and in following the location stated in the sentence. This challenge has established the first benchmark for grounding report findings in 3D CT and substantially advanced the state of the art, while making clear where further progress is needed. The test-set masks remain withheld behind a public leaderboard (`https://rexrank.ai/ReXGroundingCT`), so that new methods can be compared directly with the challenge entries.

## 2. Challenge design

The ReXGrounding Challenge was held as part of MICCAI 2026 and hosted on the ReXrank leaderboard platform (Zhang et al., 2024) (`https://rexrank.ai/ReXGroundingCT`). Its design was registered in advance with MICCAI (Baharoon et al., 2026b). The organizers, timeline, submission procedure and policies are described in Appendix A.

### *2.1. Task definition*

*Task.* Given a non-contrast chest CT volume and a free-text description of a single finding, produce a 3D binary segmentation of all regions described by that finding. Each finding receives one mask, and when a finding has several instances, such as multiple nodules, all of them are segmented together in that mask. Instances are recovered afterwards as connected components for the instance-level metrics (Section 2.4).

*Assessment aim.* The challenge assessed how well each finding described in the text was localized and delineated in 3D. The primary metric was Dice, the overlap between the predicted and reference masks of each finding, which reflects both whether the finding was found and how well its extent was captured. The secondary metrics were the hit rate, which measures whether a model localized each finding at all regardless of how precisely it outlined it, and instance F1, which measures whether it found each lesion of findings made up of several lesions. All metrics are defined in Section 2.4.

### *2.2. Challenge data*

*Data source.* Scans and reports come from CT-RATE, a public dataset of de-identified non-contrast chest CT scans and their radiology reports acquired during routine care at Istanbul Medipol University Mega Hospital, Turkey (Hamamci et al., 2026b). The original Turkish reports had been machine-translated into English and corrected by bilingual medical students in CT-RATE. For ReXGroundingCT, GPT-4 rewrote each report into standard US radiology terminology and split it into individual findings, and GPT-4o assigned each finding to a category (Baharoon et al., 2026a). A board-certified radiologist reviewed both steps on random samples. All 50 rewritten reports were judged acceptable, extraction had a false-positive rate of 0 and a false-negative rate of 0.01, and categorization of lung and pleural findings reached a precision and recall of 0.99. The finding sentences used in the challenge are the outputs of this pipeline and were not edited further.

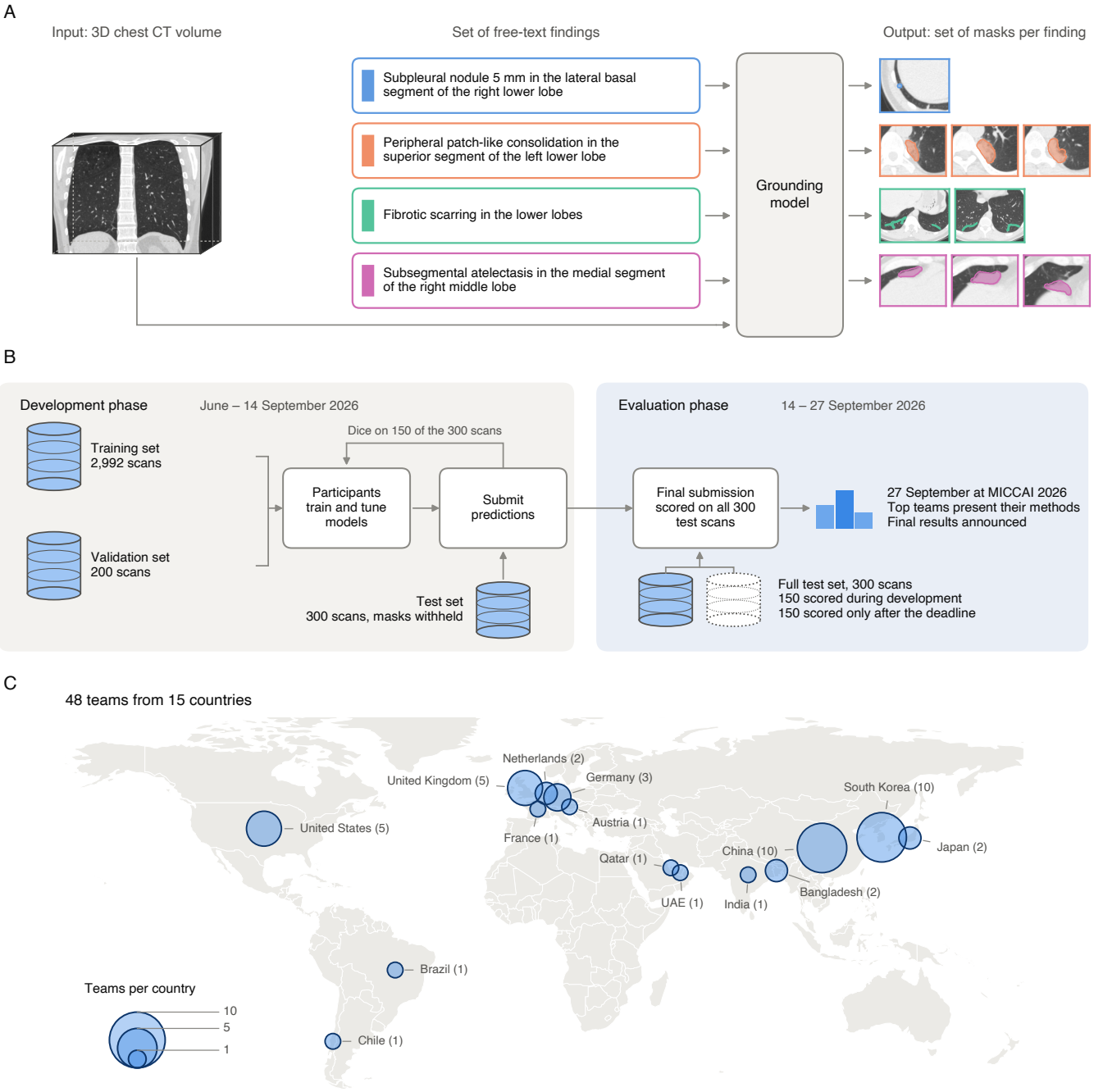



Figure 1: Overview of the ReXGrounding Challenge. **A)** The task: given a non-contrast chest CT volume and one free-text finding from its report, a model must segment that finding in 3D. Each sentence is a separate query against the same volume, and the shown masks are the radiologist reference annotations of a test case (scan from CT-RATE (Hamamci et al., 2026b)). **B)** Challenge workflow. During the development phase, teams trained on the ReXGroundingCT training set (2,992 scans with partial-instance masks) and the validation set (200 scans with exhaustive radiologist masks), and submitted predictions for the test set (300 scans, all visible to participants, with reference masks withheld), and the leaderboard reported scores on a fixed subset of 150 scans. After the deadline, each team's final submission was scored on all 300 scans and the final ranking was announced at MICCAI 2026. **C)** Participating teams by country (48 teams from 15 countries). Circle area is proportional to the number of teams. Map lines are from Natural Earth (naturalearthdata.com, version 5.1.1) and do not necessarily depict accepted national boundaries.

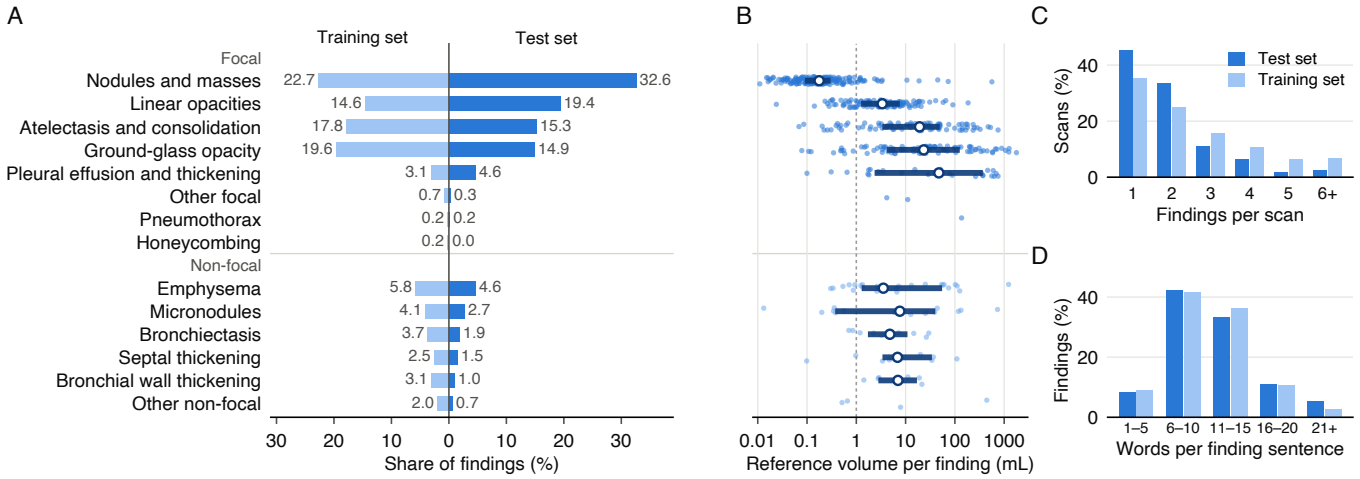



Figure 2: Challenge data. A) Finding categories as a share of all findings in the training set (left, 7,687 findings in 2,992 scans) and the test set (right, 582 findings in 300 scans), grouped into focal and non-focal categories. Numbers give the share in percent. B) Reference volume of every test-set finding by category (dots). For categories with at least five findings, the bar spans the interquartile range and the open circle marks the median. The dashed line marks 1 mL. C) Number of findings per scan. D) Length of the finding sentences. In C and D, dark bars show the test set and light bars the training set.

*Data splits.* A case is one CT volume with its set of findings. The training set contains 2,992 scans (7,687 findings) with partial-instance annotations (up to three instances per finding). The validation set (200 scans, 381 findings) and the test set (300 scans, 582 findings) were exhaustively annotated by board-certified radiologists. During the challenge, the leaderboard showed results on a fixed random half of the test set (150 scans, 280 findings), and the final ranking used all 300 scans. Figure 2 summarizes the composition of the training and test sets. Most findings in both are focal, and the reference masks span five orders of magnitude in volume, from sub-millilitre nodules to effusions and diffuse ground-glass opacities of more than a litre. The training set is identical to that of the original ReXGroundingCT release (Baharoon et al., 2026a). The challenge validation set contains the 50 original validation scans plus 150 new scans, and the challenge test set contains the 100 original test scans plus 200 new scans. No scan appears in more than one split. The new scans were annotated with the same exhaustive protocol as the original ones, by an additional group of annotators.

*Annotation.* All masks were drawn on a cloud-based, HIPAA-compliant annotation platform with brush, polygon and region-growing tools, adjustable window settings and free navigation across slices (Baharoon et al., 2026a). Training scans followed a partial-instance protocol, in which at most three representative instances of each finding were segmented. 1,400 training scans were annotated by professional annotators and accepted, corrected or returned by board-certified radiologists, and 1,592 by trained medical students supervised by radiologists until they could annotate independently. Validation and test scans followed an exhaustive protocol, in which every visible instance was segmented. The original 150 validation and test scans were annotated by four board-certified radiologists, and the 350 scans added for the challenge by an additional group of annotators under the same protocol. Findings that could not be localized on the scan, were too diffuse to outline (for example, mosaic attenuation throughout the lungs), or described normal anatomy were excluded.

*Pre-processing.* No pre-processing was applied. Scans were distributed as NIfTI volumes in their native resolution, orientation and Hounsfield units, and the reference masks share the voxel grid of their scan. Participants chose their own resampling, cropping and intensity normalization.

*Sources of error.* The main source of error in the reference standard is variability between annotators. In an inter-rater study on 50 validation and test scans (96 findings) annotated twice by non-overlapping groups of radiologists, agreement reached a Dice of 0.32, a hit rate of 0.76 and an instance F1 of 0.47 (Baharoon et al., 2026a). For focal findings, agreement reached a Dice of 0.34 and a hit rate of 0.84. For non-focal findings it was lower (Dice 0.15, hit rate 0.41), although with only 17 such findings in the study this estimate is imprecise. The lower agreement is consistent with the extent of a diffuse abnormality being inherently ambiguous. Errors can also enter through the reports themselves: translation from Turkish, the LLM rewriting and extraction steps, and occasional mismatches between a sentence and the scan. One participating team reported correcting more than 100 training annotations, including masks on the wrong side and findings absent from the source report (Section 3.2). Finally, the training and evaluation sets were annotated under different protocols. Training masks cover at most three instances of each finding, so further instances are unlabelled and appear as background, whereas the evaluation masks include every instance. A model trained on the partial masks can therefore learn to leave some instances unsegmented and is penalized for missing them at test time.

*Cohort.* The target cohort is adult patients undergoing non-contrast chest CT in routine clinical practice, and the challenge

cohort was drawn from this population. Test-set patients had a median age of 41 years (interquartile range 31–51, range 18–89) and 42% were female, similar to the training set (median 40 years, 44% female) and the validation set (median 37 years, 44% female). Test scans were acquired between 2016 and 2022 on six scanner models from Philips, Siemens and Philips–Neusoft, with a median slice spacing of 1.0 mm (range 0.75–2.0) and a median in-plane spacing of 0.71 mm (range 0.39–0.98). Each scan covers the full chest, and the segmentation targets are the lung and pleural findings described in its report, grouped into 14 categories spanning focal and non-focal patterns (Figure 2). No clinical information beyond age and sex was provided.

### *2.3. Challenge rules*

All predictions on the test set had to be fully automatic, without manual editing or case-specific tuning. Participants could use any public or private training data, pretrained models, and additional annotations, provided these were described in their method submission. Training on test cases, including self-supervised training, was not allowed. The challenge had two tracks, which differed in how the finding text could be processed. In the Main track (free-text grounding), methods had to take the finding text exactly as provided, optionally supplemented by its category label. The Overall track also admitted methods that first converted the text into a structured representation, for example by extracting location, morphology or size with a language model or parser, and every eligible Main-track submission was also ranked in the Overall track. In both tracks, each mask had to be derived from the text of its finding: methods that segmented only by category, ignoring the rest of the sentence, were not allowed. A method counted as text-conditioned if the mask it produced for a finding could depend on the wording of the sentence beyond its category, so that two findings of the same category with different descriptions, for example in different lobes, could receive different masks. A fixed-class model that maps each category label to a single mask was therefore not allowed, even though the category labels were provided. Methods could use anatomical segmentations as additional inputs and post-process masks, for example by restricting them to the anatomical region named in the finding. All six best-performing teams submitted to the Main track, so the two tracks produced the same winners, and we report a single ranking.

### *2.4. Assessment method*

*Metrics.* The primary metric was Dice, computed for each finding over its whole mask, averaged over the findings in each case, and then averaged over cases. Before any metric was computed, predicted connected components smaller than 10 voxels were removed and at most the 50 largest components per finding were kept. An empty prediction for a finding therefore scored a Dice of zero. We defined the hit rate as the proportion of findings with a Dice of at least 0.1. For the instance-level metrics, the reference and filtered predicted masks were split into connected components, which were matched one to one in order of decreasing Dice, and a match counted as a true positive when its Dice was at least 0.2. Instance precision, recall and F1 follow from these matches. Mean Dice per finding and mean Dice over matched instances were also reported but not used for ranking.

Dice was used as the single primary metric so that all findings could be ranked on a common scale, and it was computed over the whole mask of each finding, which avoids defining instances for diffuse patterns where boundaries between lesions are arbitrary. As recommended by Metrics Reloaded (Maier-Hein et al., 2024), Dice was complemented by the detection metrics hit rate and instance F1. Averaging per case before averaging over cases gives every scan equal weight, so scans with many findings do not dominate. Dice is, however, a poor fit for diffuse findings, whose extent is inherently ambiguous (Section 2.2). Results for these findings should therefore be read alongside the hit rate, and we return to this in the Discussion.

*Ranking.* For each team we ranked the submission it declared as final or, if none was declared, its best-scoring submission on the public half of the test set. 8 of the 48 leaderboard entries (teams) declared a final submission, and the remaining 40 defaulted to their best public-half submission. Teams were ranked by the primary metric computed on the full 300-scan test set, following an aggregate-then-rank scheme: each team's Dice was first averaged over all findings and cases, and teams were then ordered by this mean. Missing cases could not occur in the ranking, because a submission was scored only if, for each of the 300 test scans, it contained one mask per finding with the same dimensions as the scan. No two teams had the same mean Dice, so there were no ties. We chose aggregate-then-rank because it reflects how much better one method is than another, whereas ranking teams on each scan and averaging their ranks (rank-then-aggregate) reflects only how often it is better. As a sensitivity analysis, we also ranked the teams with rank-then-aggregate (Section 3.4).

*Statistical analysis.* The precision of each team's mean Dice, hit rate and instance F1 was estimated with 1,000 bootstrap replicates that resampled the 300 test cases with replacement and recomputed each score with the same aggregation as the ranking. The resulting 95% percentile confidence intervals are given in Table 1 and Table C.1. Ranking variability was assessed by re-ranking all teams in each replicate and reporting the distribution of each team's rank and the range covering 95% of its bootstrap ranks (Figure C.1A), and agreement between bootstrap and observed rankings was summarized with Kendall's $\tau$ (Section 3.4) (Maier-Hein et al., 2018; Wiesenfarth et al., 2021). For each pair among the ten best teams, the proportion of replicates in which the higher-ranked team scored higher is reported in Figure C.1B, together with whether it scored significantly higher in a one-sided paired Wilcoxon signed-rank test on per-case Dice, with Holm correction for the 45 comparisons. All analyses used Python with NumPy and SciPy.

*Controlled prompt experiment.* To test how the location, size and other details given in a finding sentence affect the predicted

mask, the final methods of the best teams were run on edited versions of the test sentences. Each edit changed one aspect of a sentence and left the scan and the reference mask unchanged. Information was removed by deleting the stated size, the lobe or segment, or the side, or by keeping only the name of the finding. Wrong information was introduced by swapping left and right, naming a different lobe in the same lung, or changing a small stated size to 30 mm and a large one to 3 mm. Findings in both lungs were also prompted once for each side. Size and side edits were made with rules, and the other edits were made by a large language model (Claude Opus 5.5, Anthropic). Each edit was evaluated only on the findings it changed, against the same team's result for the original sentence, using the paired difference in Dice with a 95% bootstrap confidence interval over scans and a two-sided Wilcoxon signed-rank test. For the location edits we also measured the share of the predicted mask that lay on each side of the body. The effect of each edit on Dice and on the location of the mask is reported in Section 3.7 and summarized in Figure 6C, and Figure D.1 shows example masks predicted for original and edited sentences.

## 3. Results

Throughout this paper, teams are numbered by their final rank, so Team 1 is the winning team and Team 10 the tenth-placed team (Table 1).

### *3.1. Participation*

Submissions came from 53 team registrations. Some groups had registered more than once, so we merged registrations that shared team members or institutional contact details into a single team, keeping that team's best-scoring final submission. After merging, 48 teams from 15 countries had made 760 submissions in total (median 4 per team). The largest contingents came from China (10 teams, including one from Hong Kong and one joint China–USA team) and South Korea (10) (Figure 1C). We classified each team by the affiliations it listed at registration for the team and its members. 35 of the 48 teams listed only universities, research institutes or hospitals, 6 listed only companies, and 4 listed both a company and a university or hospital. Industry involvement was concentrated at the top of the ranking: five of the six best-performing teams included a company, and the only one without industry involvement was Team 3. Submissions were concentrated at the end of the development phase: the first submission to appear on the leaderboard arrived on 5 July, three weeks after the leaderboard opened, and 74% of all submissions were made in the final two weeks (Figure 4A). The best score on the public half rose from 0.28 for the first submission to 0.386 on 13 September, the day before the deadline (Figure 4B).

### *3.2. Overview of the top-performing methods*

All three winning teams followed the same recipe. A frozen Qwen3-Embedding text encoder (Zhang et al., 2025b) turns the finding sentence into a single embedding vector, and a 3D segmentation network takes the CT volume together with this vector and outputs the mask of that finding. The teams differed in

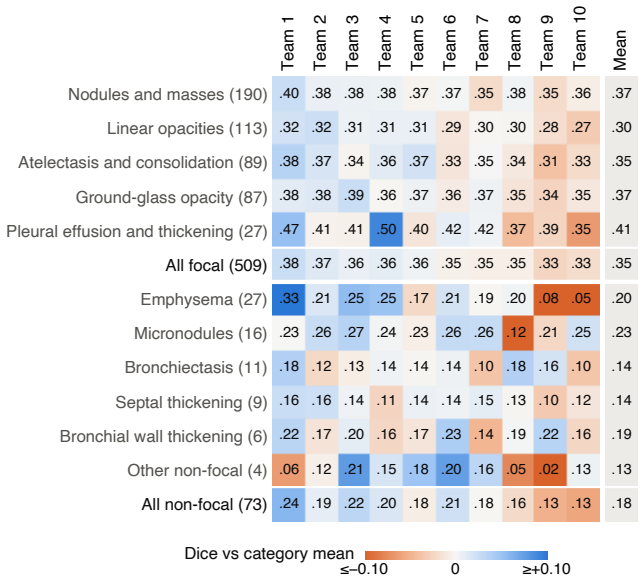



Figure 3: Mean Dice per finding of the ten best teams by finding category on the full test set. Numbers in cells give the Dice, and colour shows how far each team is above (blue) or below (orange) the mean of the ten teams for that category. The last column gives that mean. Numbers in brackets give the number of test findings. Pneumothorax (1 finding) and other focal findings (2) are included in the focal total but not shown.

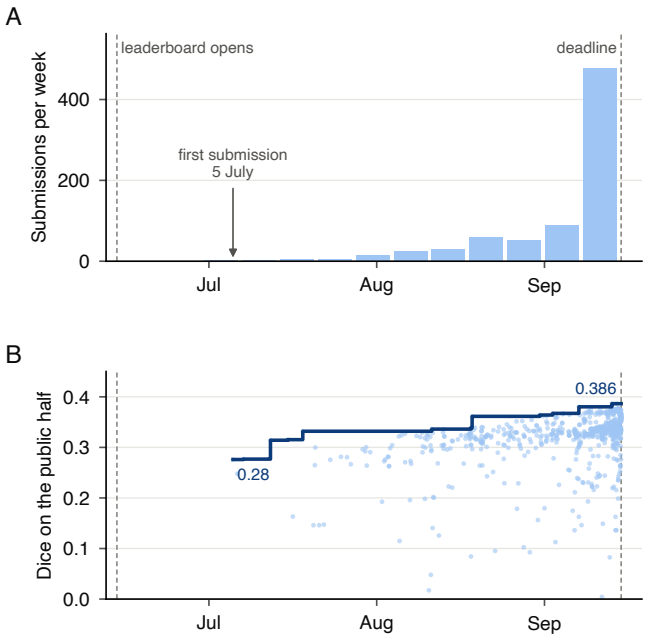



Figure 4: Submissions over the development phase. A) Scored submissions per week, in seven-day windows ending at the deadline. B) Dice on the public half of the test set for every scored submission (dots) and the best score reached so far (line). Dashed lines mark the opening of the leaderboard and the submission deadline.

Table 1: Final ranking on the full 300-scan test set (top 10). Teams are numbered by their final rank. CI, 95% bootstrap confidence interval over the test scans (1,000 replicates, Section 3.4). The full leaderboard is in Appendix C.

| Team | Dice [95% CI] | Hit rate [95% CI] | Instance F1 [95% CI] |
|---|---|---|---|
| Team 1 | 0.3755 [0.352–0.399] | 0.7680 [0.732–0.802] | 0.3410 [0.312–0.369] |
| Team 2 | 0.3654 [0.344–0.387] | 0.7629 [0.730–0.797] | 0.3428 [0.316–0.371] |
| Team 3 | 0.3621 [0.342–0.383] | 0.7646 [0.737–0.795] | 0.3336 [0.306–0.364] |
| Team 4 | 0.3605 [0.339–0.382] | 0.7766 [0.744–0.809] | 0.3116 [0.285–0.341] |
| Team 5 | 0.3526 [0.330–0.375] | 0.7457 [0.713–0.781] | 0.1899 [0.166–0.216] |
| Team 6 | 0.3519 [0.332–0.373] | 0.7646 [0.732–0.799] | 0.2435 [0.217–0.271] |
| Team 7 | 0.3432 [0.322–0.366] | 0.7526 [0.721–0.785] | 0.2611 [0.238–0.288] |
| Team 8 | 0.3393 [0.317–0.362] | 0.7423 [0.708–0.777] | 0.2463 [0.220–0.275] |
| Team 9 | 0.3329 [0.313–0.354] | 0.7165 [0.683–0.750] | 0.2481 [0.222–0.278] |
| Team 10 | 0.3329 [0.311–0.355] | 0.6890 [0.654–0.726] | 0.2985 [0.269–0.328] |

three places. The first is the image encoder, which was the chest CT foundation model CIRCLE for Team 1, a residual-encoder nnU-Net trained from scratch for Team 2, and the VoxTell encoder together with TAP-CT (Veenboer et al., 2025) for Team 3. The second is how the text enters the network. Teams 1 and 3 kept the VoxTell decoder (Rokuss et al., 2026), in which the text embedding acts as the query of a small transformer that cross-attends to the image features at the bottleneck. Its output is projected to every decoder stage, where it is multiplied channel-wise with the image features and the product is concatenated back onto them, so the text steers the mask at every resolution. Team 2, and Team 3's TAP-CT branch, used FiLM (Perez et al., 2018) instead, in which small networks map the embedding to a per-channel scale and shift that are applied to the image features, with no cross-attention between text and image. The third is everything around the model. Team 1 added 950 in-house annotated scans and averaged checkpoints, Team 2 used only the challenge data and averaged 13 models, and Team 3 averaged 17 models and post-processed each mask by restricting it to the lobe named in the sentence, choosing a threshold per finding and removing components predicted to lower Dice. More detailed descriptions of each method are given in Appendix B.

*Team 1.* The winning entry paired the image encoder of CIRCLE, a chest CT foundation model pretrained on 403,500 non-contrast scans with paired reports from 288,740 patients through image–report contrastive learning and supervised classification of 37 abnormalities, with the text-prompted segmentation framework of VoxTell (Rokuss et al., 2026). The image encoder and the VoxTell decoder were fine-tuned on the challenge training set, and the text encoder was kept frozen. Scans were resampled to $0.65 \times 0.65 \times 1.15$ mm and trained on $384 \times 320 \times 224$ voxel inputs with a Dice–cross-entropy (DiceCE) loss. On a 50-case validation subset, pretraining raised mean Dice from 0.277 (same architecture from scratch) to 0.327, and adding in-house data (800 pulmonary-nodule segmentations and 150 cases of less common thoracic disease) raised it further to 0.356. The final submission averaged checkpoints of a single model from several epochs.

*Team 2.* The second-placed team used no external data and a deliberately simple architecture they called Grounder: a 3D nnU-Net with the residual-encoder "L" configuration (Isensee et al., 2021, 2024), conditioned on the finding text through feature-wise linear modulation (Perez et al., 2018). The sentence was embedded with Qwen3-Embedding-4B (Zhang et al., 2025b), and a two-layer MLP mapped the embedding to scale and shift parameters applied at every decoder stage. Two choices drove most of the gain. First, a large input patch of $448 \times 448 \times 256$ voxels gave the network enough anatomical context to resolve location phrases. Second, in-distribution text augmentation replaced each finding sentence, with some probability, by a real sentence from another training case that described the same finding type at the same location. The final submission averaged 13 Grounder models trained with different patch sizes, which the team found to be complementary. Adding lung-lobe masks from TotalSegmentator as input channels, an explicit finding-category embedding, and negative text examples did not help.

*Team 3.* The third-placed team combined two model families. The main branch was an ensemble of 12 VoxTell models, in which the text embedding (Qwen3-Embedding-4B) enters an nnU-Net decoder as a query through cross-attention, trained on $192^3$ patches. A second branch, TAP-CT (Veenboer et al., 2025), a 3D vision transformer pretrained in the style of DINOv2 (Oquab et al., 2023) with a convolutional skip decoder and FiLM conditioning on Qwen3-Embedding-0.6B, was weaker alone but added diversity, and the two were fused with weights of 0.8 and 0.2. All members were trained on the training and validation sets with synonym-based text augmentation (31 synonym groups), but their recipes otherwise varied. Two of the 12 VoxTell members added 8,000 CT-RATE sentence–mask pairs with pseudo-masks of pleural effusions and nodules from TotalSegmentator (Wasserthal et al., 2023), one added lobe segmentation as an auxiliary task, and five oversampled the validation set five-fold late in training. Post-processing had four steps. Each mask was restricted to the lobe named in the sentence (mapped offline with an LLM and segmented with TotalSegmentator), or to the named lung when no lobe was given. A learned model then set a separate threshold for each finding, a small 3D CNN removed components predicted to lower Dice, and voxels claimed by several findings went to the most confident one. The team found the lobe restriction, the per-finding thresholds and TAP-CT as a second architecture to be the most helpful.

*Team 4.* The fourth-placed team also fine-tuned VoxTell, on $288 \times 288 \times 192$ crops at $0.75 \times 0.75 \times 1.25$ mm, and reported that fine-tuning beat training the same architecture from scratch by more than 0.05 Dice. Three further steps improved performance. They audited the training annotations and corrected more than 100 cases, including masks on the wrong side, drawing artifacts, and findings that the source report did not contain. They extracted 62,705 finding sentences from 18,652 CT-RATE reports with MedGemma (Sellergren et al., 2026) and used these scans for mean-teacher semi-supervised train-

ing, together with external segmentation datasets and a presence loss on pooled voxel logits. Finally, each model was fine-tuned for a further 500 steps on the validation set alone, whose exhaustive annotation protocol is closest to the test set. As a post-processing step, predictions for sentences naming a side were trimmed to that lung. The team reported no gain from a Tversky loss intended to tolerate partial annotations, and no improvement on bronchiectasis and micronodules, which they attributed to annotation variability. Their entry had the highest hit rate of all teams (0.777).

*Teams 5 and 6.* The next two teams were within 0.01 Dice of fourth place. Team 5 combined two VoxTell-based 3D models with a model that segments native-resolution axial slices. Both 3D models also received a coarse map of where the finding is likely to lie, from a vision–language model the team pretrained on CT-RATE. Because the training masks cover at most three instances of a finding, the team excluded uncertain lung voxels from the loss rather than treating them as background, and at inference an LLM was used to trim each mask to the lobe or side named in the sentence. Team 6 extended VoxTell with several queries formed from the individual words of the sentence, weighted by the whole-sentence embedding, and trained with a loss that penalises predictions inside a finding of another category in the same scan. They used specialist models for selected categories such as nodules, and removed predictions outside the named lobe.

### 3.3. Final ranking

Dice across all teams ranged from 0.146 to 0.375 (median 0.280). The top four teams lay within 0.015 Dice, and 3rd and 4th place were separated by 0.0016. The overall ranking also masks large differences between finding categories, which are shown in Figure 3.

*Comparison with models before the challenge.* Before the challenge, the public ReXGroundingCT leaderboard evaluated models on a 100-scan subset of the challenge test set (Baharoon et al., 2026a). General text-promptable segmentation models, which are trained on names of anatomical structures and lesion classes rather than on descriptive findings, failed out of the box. SAT (Zhao et al., 2025c), SegVol (Du et al., 2024) and BiomedParse-V (Zhao et al., 2025b) all reached a Dice close to 0. Fine-tuning SAT on the ReXGroundingCT training set raised its Dice to 0.205 (hit rate 0.473), and the best entry at the time, VoxTell (Rokuss et al., 2026), reached 0.285 (hit rate 0.615). On the same 100 scans, the winning team reached a Dice of 0.390 (hit rate 0.712), and 23 of the 48 teams scored above VoxTell.

### 3.4. Ranking stability

The ranking on the public half of the test set closely matched the final ranking on all 300 scans. The first two places were unchanged, and places three to eight changed only by swaps between neighbouring pairs. The ranking was stable overall (Figure C.1): bootstrap rankings agreed closely with the observed one (median Kendall's $\tau = 0.95$, interquartile range 0.94–0.96). At the top, however, the margins were within the uncertainty of the test set. Team 1 ranked first in 89% of bootstrap replicates and within the top three in 99%, but the 95% confidence intervals of the top four teams overlapped (Table 1), and none of the differences among them was significant after correction. Team 1 scored higher than Team 2 in 91% of replicates, whereas Team 3 beat Team 4 in only 57%. The winner was significantly better than the fifth- and sixth-placed teams (adjusted $p = 0.017$ and 0.009). Ranking with rank-then-aggregate instead of the official aggregate-then-rank scheme (Section 2.4) gave a similar ranking overall (Kendall's $\tau = 0.96$) and the same four teams at the top, but in a different order: Team 2 ranked first, Team 3 second and Team 1 third, with Team 4 fourth. Team 1 had the highest mean Dice, whereas Team 2 had the best average rank across scans, so the choice of scheme decided the winner among teams whose scores were statistically indistinguishable.

### 3.5. Performance by finding type

Focal findings were segmented far better than non-focal findings by all top teams: across the six best teams, Dice on focal findings (n=509) ranged from 0.348 to 0.378 with hit rates of 0.78–0.82, whereas Dice on non-focal findings (n=73) ranged from 0.176 to 0.243 with hit rates of 0.47–0.53. Focal performance was therefore tightly clustered, and the winner's margin came largely from non-focal findings (0.243 vs. at most 0.216 for the other top-six teams). Figure 3 breaks these results down by category. Among focal findings, pleural effusion and thickening was segmented best (Dice 0.35–0.50) and linear opacities worst (0.27–0.32). Non-focal categories varied far more between teams than focal ones. The winner's non-focal lead came mainly from emphysema, where Team 1 reached 0.33 and no other top-ten team exceeded 0.25. Bronchiectasis and septal thickening were hard for every team (Dice at most 0.18), and the smallest non-focal categories, with 4 to 9 findings each, were too small to rank teams reliably.

### 3.6. Qualitative comparison

Figures 5 and 7 show the final submissions of the three winning teams on five findings that were grounded well and five that were not, chosen to cover different finding types. A consolidation in the left lower lobe and a 5 mm subpleural nodule were localized by all three teams (Dice 0.58–0.82), with differences mainly in boundary detail. The teams' solutions also followed longer, more detailed sentences. For consolidations with air bronchograms and ground-glass areas in the right upper lobe, most prominent in the posterior segment, all three reached a Dice above 0.80. For subsegmental atelectasis at three named sites in both lungs, Teams 1 and 3 placed a mask at every named site and Team 2 at two of them (Dice 0.49–0.66). Other findings separated the teams. Team 1 segmented an air cyst in the lingula almost completely (Dice 0.85), whereas Teams 2 and 3 marked only part of it.

The failures show several distinct problems. Team 1 was the only team to capture a 10 cm mass in the right upper lobe (Dice 0.71), although its mask extended beyond the reference, while Team 2 missed it entirely. For focal consolidation described

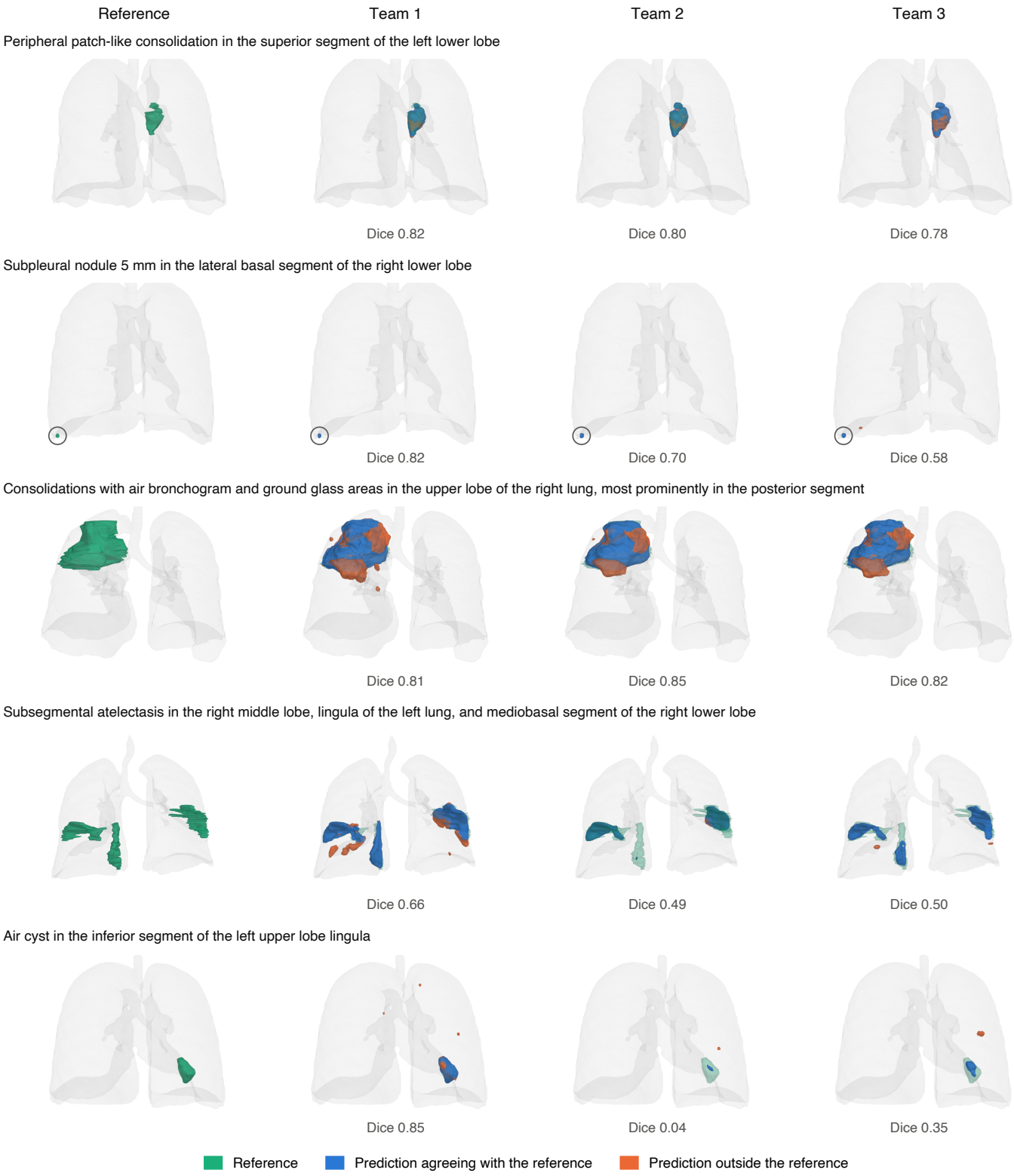



Figure 5: Examples of successful grounding by the three winning teams. Masks are rendered in 3D inside the lungs and viewed from the front (patient's right on the left). The reference is shown in green, solid in the first column and translucent in the others. Predicted surfaces are blue where they lie within 3 mm of the reference and orange elsewhere. Lesions smaller than 1 mL are circled. Dice is computed over the whole 3D mask of the finding.

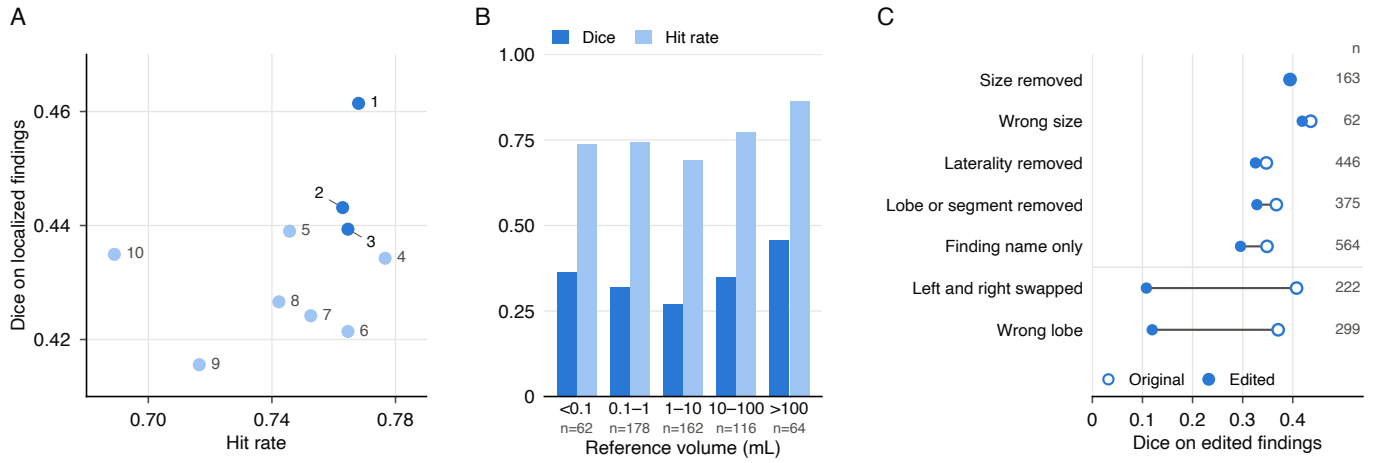



Figure 6: Further analyses. A) Hit rate against mean Dice on localized findings (Dice ≥ 0.1) for the ten best teams. Numbers give the final rank. The three winning teams are shown in darker blue. B) Dice (dark) and hit rate (light) by reference volume of the finding, pooled over the ten best teams. C) Controlled prompt experiment. Dice on the findings changed by each edit, with the original sentence (open) and the edited sentence (filled), averaged over Teams 1 to 4. Numbers give the number of edited findings.

in the lateral segment of the right middle lobe, Teams 1 and 2 instead segmented a consolidation in the right upper lobe that the same report lists as a separate finding (Dice 0.19 and 0.46 against that finding), so the models recognized the abnormality but not the location named in the sentence. For subsegmental atelectasis in the medial segment of the right middle lobe, all three teams segmented a thin neighbouring structure rather than the atelectatic segment (Dice at most 0.02). Calcified pleural plaques were found only in part by Team 1 (Dice 0.52) and largely missed by the other two teams. For minimal emphysema, a non-focal finding, Team 1 predicted nothing and the other two teams marked regions away from the annotated foci.

### *3.7. Further analyses*

Three further analyses examine what separated the teams and which findings were hard. First, detection and delineation. Across all 48 teams, hit rate and Dice were closely related (Spearman $\rho = 0.97$), so teams that localized more findings also segmented them better. Among the ten best teams, however, hit rates lay within a narrow band (0.69–0.78) and the winner's margin came from delineation: Team 1 reached a mean Dice of 0.46 on the findings it localized, against at most 0.44 for any other team (Figure 6A). Team 4 had the highest hit rate of all teams (0.78) but a lower Dice on localized findings (0.43), which left it fourth.

Second, finding size. Pooling the ten best teams, performance was not worst on the smallest findings. Nodules below 0.1 mL were localized in 78% of cases with a mean Dice of 0.39, similar to nodules of 0.1–1 mL (85%, 0.38). The lowest scores were for findings of 1–10 mL (Dice 0.27, hit rate 0.69), a range dominated by linear opacities, and the highest for findings above 100 mL (0.46, 0.86), mostly ground-glass opacities, consolidations and pleural effusions (Figure 6B). Size and finding type are therefore entangled, and the dip reflects the difficulty of thin linear opacities more than volume itself.

Third, language. Teams 1 to 4 ran their final methods on edited sentences (Section 2.4, Figure 6C). Teams 2, 3 and 4 responded to the edits in the same way, and Team 1 differed in the ways noted below. Removing the stated size changed Dice by less than 0.01 for all four teams (163 findings). For Teams 2, 3 and 4, removing the side lowered Dice by 0.02–0.03, removing the lobe or segment by 0.04–0.06, and keeping only the name of the finding by 0.05–0.08. Team 1 lost at most 0.02 under these edits and thus relied less on the stated location. Wrong locations misled all four models. With left and right swapped, Dice fell from 0.40–0.42 to 0.08–0.14 and the hit rate from 0.77–0.80 to 0.16–0.31. For Teams 2, 3 and 4, the share of the predicted mask on the side of the finding fell from 92–93% to 22–24%, whereas Team 1 more often returned an empty mask (31% of findings, against 12–21%). Naming the wrong lobe in the same lung lowered Dice from 0.37–0.38 to 0.07–0.17 (Figure D.1). A wrong size left the masks of Teams 2, 3 and 4 almost unchanged (Dice changed by less than 0.01), whereas the Dice of Team 1 fell from 0.46 to 0.41 on these 62 findings (Wilcoxon $p = 0.44$). When a finding in both lungs was prompted for one side only, at least 85% of the predicted mask lay on the named side, and the union of the two masks scored within 0.01 of the original sentence. The models therefore relied on the stated side and lobe but hardly on the stated size.

## 4. Discussion

### *4.1. Interpretation of results*

The ReXGrounding Challenge was organized to advance the 3D grounding of radiology report findings. Such grounding is a step towards tools that let clinicians see, measure and track each reported finding in the image. It is also a step towards the ultimate goal of grounded radiology report generation, in which AI systems pair each finding they generate with its location in the volume so that their output can be verified.

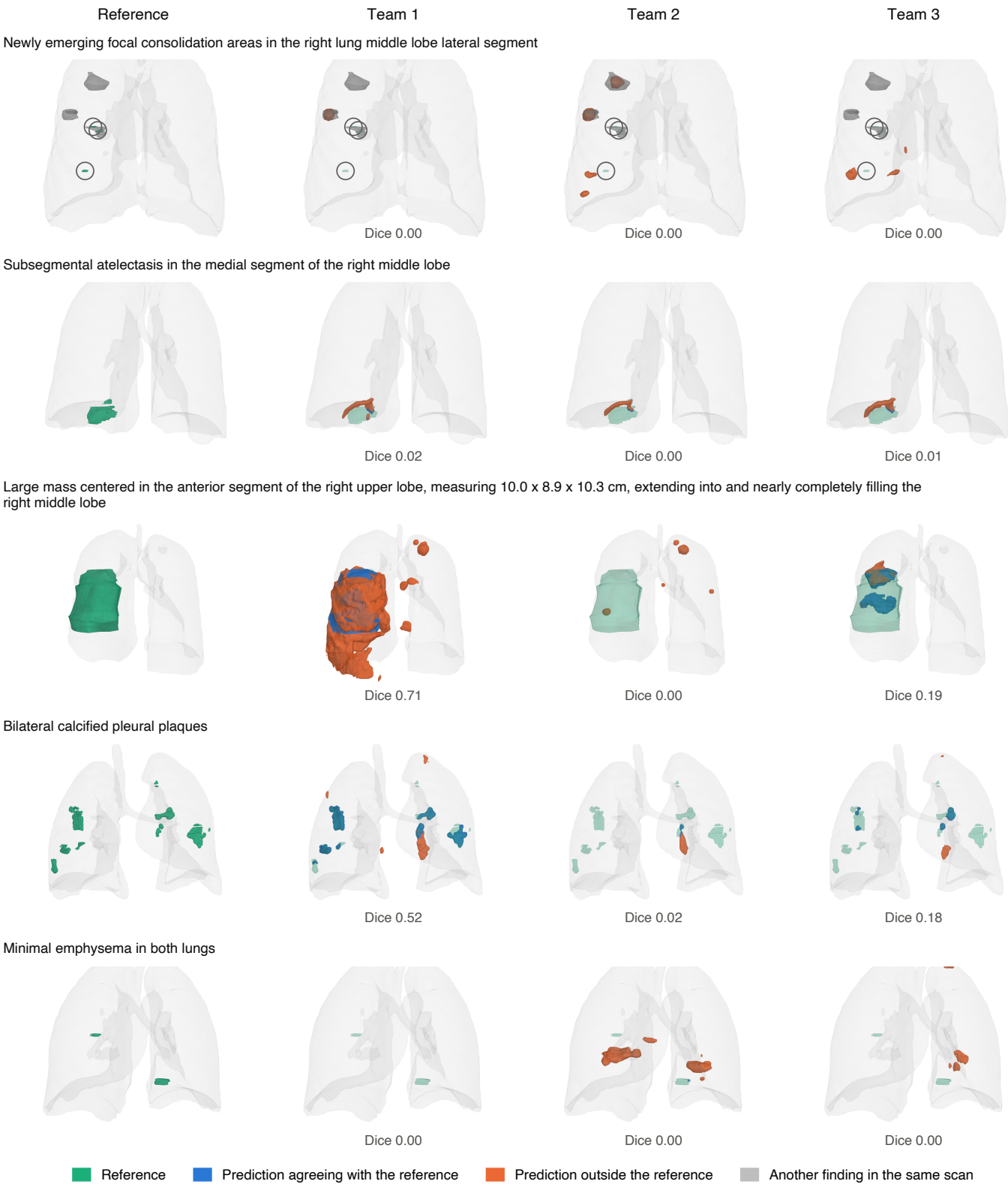



Figure 7: Examples of grounding failures by the three winning teams, shown as in Figure 5. In the first row, other reference findings of the same scan are shown in grey: two of the teams segmented a consolidation that the report describes as a separate finding in the right upper lobe.

The challenge substantially advanced the state of the art. On the 100 scans of the pre-challenge benchmark, the winning team reached a Dice of 0.390, against 0.285 for the best model before the challenge. On the full test set, the best entries reached a Dice of 0.375 and found up to 78% of findings. Performance depended strongly on the type of finding. Focal findings with a clear boundary, such as nodules, consolidation and pleural effusion, were segmented far better than diffuse non-focal findings, and the winner's margin came mostly from non-focal findings, in particular emphysema (Figure 3).

Measured by Dice and hit rate, the best methods now match the agreement between two independent groups of radiologists, who reached a Dice of 0.32 and a hit rate of 0.76 when annotating the same 50 scans (Section 2.2). The same holds for focal findings: focal Dice of the top teams (0.35–0.38) is at or above inter-reader agreement on focal findings (0.34), and non-focal Dice (0.18–0.24) is above the 0.15 reached by radiologists, although that figure rests on only 17 non-focal findings. The comparison is approximate, because the inter-reader study used a different set of scans, but it suggests that Dice against a single reference is close to saturating against annotation variability.

The task is nonetheless not solved. Instance F1 remained well below inter-reader agreement (0.34 for the best team against 0.47), so models often found part of a multi-instance finding but not every lesion, and findings with four or more instances had a lower Dice than single-instance findings (Section 3.7). The top teams also localized fewer focal findings than radiologists did (hit rate 0.78–0.82 against 0.84). Finally, the qualitative failures show that models sometimes segmented the right abnormality at a location other than the one stated in the sentence, which is the failure that grounding is meant to prevent and which an overall Dice does not reveal.

For radiology, the results show that grounding a report sentence in 3D is now feasible for focal findings, which make up most of the test set. A model that localizes a reported nodule or consolidation could link each sentence of a report to the image, so that referring physicians can find what the report describes, and could support measurement and comparison over time. The current level of performance is not sufficient for unsupervised clinical use, in particular for diffuse findings and for findings described at a specific location, but it is sufficient for tools that propose a region for a reader to confirm. For method development, the challenge provides the first public benchmark of this task at scale, with exhaustive radiologist annotation and a public leaderboard that keeps the test masks hidden, so that new methods can be compared directly with the entries reported here.

### *4.2. Limitations and future work*

The challenge has several limitations. All scans come from a single source, CT-RATE, and are non-contrast chest CT acquired at one centre, so the results may not transfer to other institutions, protocols or body regions. The test set is dominated by focal findings, non-focal findings make up only 12.5% of it, and several categories contain fewer than ten findings, so per-category results for rare findings are imprecise. Dice is a poor fit for diffuse findings, whose extent varies even between radiologists. More broadly, agreement between radiologists was itself modest (Dice 0.32 on the same 50 scans), because the annotation protocol left room for interpretation, for example in how far a diffuse finding extends or which of several lesions a sentence refers to. A single reference mask is therefore an uncertain target, and small differences between teams are hard to interpret. The training masks cover at most three instances of each finding whereas the test masks are exhaustive, which penalizes models for instances that the training data left unlabelled. Because the test scans and their reports are publicly available in CT-RATE, models pretrained on CT-RATE may have seen the test images and reports, although never the reference masks. Finally, the rule for which submission counted differed between the challenge design document and the challenge website, and teams that had registered more than once were identified and merged only after the deadline.

A future edition could address these limitations in four ways. First, a more standardized annotation protocol, with explicit rules for the extent of diffuse findings and for which instances a sentence covers, together with several independent annotators for each test finding, would raise inter-rater agreement and allow methods to be compared against a consensus reference. Second, data from several centres, including contrast-enhanced CT, would test how well methods generalize. Third, the task could be extended beyond chest CT and beyond lung and pleural findings to other organs, body regions and imaging modalities. Fourth, it could move from grounding given findings to grounded report generation, in which a model both writes the report and segments every finding it describes.

## 5. Conclusion

The ReXGrounding Challenge was the first challenge on segmenting findings described in free-text radiology reports in 3D chest CT. The best methods, built on pretrained text-promptable models with large anatomical context, ensembling and anatomy-aware post-processing, reached the level of agreement between independent radiologists on overall Dice. Three gaps remain. Models often find only some of the lesions of a multi-instance finding, segment diffuse findings far less accurately than focal ones, and sometimes place the mask at a location other than the one the sentence names. These remaining gaps, rather than overall Dice, should guide the next generation of grounding methods. The public leaderboard at `https://rexrank.ai/ReXGroundingCT` remains open, so that new methods can be evaluated on the same hidden test set.

## Declarations

*Author contributions.* Organizers. Mohammed Baharoon: Conceptualization, Methodology, Software, Formal analysis, Visualization, Project administration, Writing (original draft). Mahmoud Hussain Alabbad, Sung Eun Kim and Xiaoman Zhang: Project administration (challenge organization). Lucas Bijnens: Project administration (annotation organization).

Luyang Luo: Project administration (challenge organization), Writing (review and editing). Pranav Rajpurkar: Supervision, Writing (review and editing). Annotation. Michael Moritz and Hassan R. Alomaish: Data curation, Supervision. Ali Piracha, Reem Alwasiah, Talal Yahyan, Abdulrahman Alamoudi, Mashael Alshehri, Rayan Alsager, Hannah Hancock, Anna M. Hunkemöller, Peter Youssef, Mona A. Alhammad, Afnan Aljumayan, Brecht Van Berkel, Eric Bijnens, Saud Alhussain, Maarten Steyvers, Rolf Symons, Bryan Eelen, Ayed Alaklubi, Maha Buraykan Alshammari, Amirah Fahad Alshammeri, Tomas Gilvydis, Ibrahim AlTurki, Monerah Almohideb, Ali Alburkani, Shahad Alhomaiani and Fatima Jabali: Data curation. Team 1. Zirong Chen: Data curation, Methodology, Software, Validation. Jiayu Wang: Formal analysis, Investigation, Methodology, Software. Xi Ouyang: Methodology, Software, Validation, Visualization. Xiaodan Ye: Data curation, Formal analysis, Investigation. Weiqiu Jin: Data curation, Formal analysis, Investigation. Mingyang Yu: Formal analysis, Investigation, Software. Dijia Wu: Formal analysis, Investigation, Methodology, Writing (review and editing). Team 2. Kwang-Hyun Uhm: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Software, Supervision, Validation. Inhwa Son: Data curation, Formal analysis, Validation. Team 3. Rachel van 't Hull: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing (original draft). Kevin Groot Lipman: Conceptualization, Methodology, Resources, Software, Supervision, Writing (review and editing). Tim Veenboer: Methodology, Resources. Cees Snoek: Supervision. Team 4. Daesung Kim: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration. Junha Im: Data curation, Formal analysis, Investigation. Youngjun Jun: Formal analysis, Investigation. Sangwon Lee: Funding acquisition (Team 4). Katelyn Morrison: Conceptualization, Supervision. Adam Perer: Funding acquisition (Team 4), Supervision. Mijin Yun: Funding acquisition (Team 4), Supervision. VoxTell. Maximilian Rokuss, Moritz Langenberg, Yannick Kirchhoff, Benjamin Hamm and Klaus Maier-Hein: Writing (review and editing).

*Conflicts of interest.* Zirong Chen, Jiayu Wang, Xi Ouyang and Dijia Wu are employees of Shanghai United Imaging Intelligence, Shanghai, China, where they work on research and development of AI algorithms for medical imaging, and Mingyang Yu was an intern there during this work. Kwang-Hyun Uhm is Chief Technology Officer of and holds equity in MEDAI Inc., a company developing medical imaging AI software. MEDAI Inc. provided no funding for this work. Daesung Kim is an employee of NewcureM Inc., a medical AI and radiopharmaceutical company, and Mijin Yun is its Chief Technology Officer. The PhD positions of Rachel van 't Hull and Tim Veenboer at the Netherlands Cancer Institute are funded by Kaiko, a company developing medical imaging foundation models, through a collaboration with the institute, and Kevin Groot Lipman is employed in part by Kaiko. Kaiko was not otherwise involved in this work. The remaining authors declare no conflicts of interest.

*Data availability.* The ReXGroundingCT training and validation data, including the findings and reference masks, and the test scans with their findings are available at `https://huggingface.co/datasets/rajpurkarlab/ReXGroundingCT` under a CC BY-NC-SA 4.0 licence. The CT scans and reports originate from CT-RATE (Hamamci et al., 2026b). The test-set reference masks remain withheld so that the public leaderboard at `https://rexrank.ai/ReXGroundingCT` can continue to evaluate new methods on the same test set.

*Code availability.* The evaluation code that produced the rankings is available with the dataset at `https://huggingface.co/datasets/rajpurkarlab/ReXGroundingCT`. The code of Team 1 is available at `https://github.com/uii-open-source/CIRCLE/`, and its model weights are not publicly released. The code of Team 2 will be released at `https://github.com/khuhm/Grounder`. The code of Team 3 is available at `https://github.com/nki-radiology/TAGS-ReXGroundingCT` under the Apache-2.0 licence, and its model weights at `https://huggingface.co/fomofo/TAGS-ReXGroundingCT` (CC BY-NC-SA 4.0 for the VoxTell models and CC BY-NC 4.0 for the TAP-CT models). The code of Team 4 is available from the team on request.

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## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Opus 5.5 and Fable 5.1, Anthropic) to help draft and edit the text, write analysis and figure code, and check reported numbers against the analysis outputs. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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## Appendix A. Challenge organization

*Name and organizers.* The ReXGrounding Challenge was held as part of MICCAI 2026. It was organized by Mohammed Baharoon, Pranav Rajpurkar, Luyang Luo, Xiaoman Zhang, and Sung Eun Kim (Harvard Medical School, USA) and Mahmoud Hussain Alabbad (King Fahad Hospital, Saudi Arabia). The challenge design was registered in advance with MICCAI and is publicly available (Baharoon et al., 2026b).

*Life cycle and platform.* The challenge was a one-time event with a fixed submission deadline of 14 September 2026, 11:59 PM ET, at which the official rankings were fixed. It was hosted on the ReXrank leaderboard platform (Zhang et al., 2024) (`https://rexrank.ai/ReXGroundingCT`), where teams registered, submitted predictions for the test set, and followed their scores on a public half of the test set during the development phase. Results were announced at a dedicated session at MICCAI 2026 in Strasbourg, France, on 27 September 2026, after which the leaderboard reopened for new submissions as an ongoing benchmark.

*Participation, awards and publication.* Members of the organizers' institutions could participate and were eligible for awards, although none took part. The six best-performing teams received certificates and presented their methods at the challenge session at MICCAI 2026 in Strasbourg, France, on 27 September 2026. Co-authorship of this paper was originally offered to the top three teams. Because the fourth-placed team finished only 0.0016 Dice behind the third, a margin too small to separate them meaningfully, we extended co-authorship to the top four teams (up to eight members per team). Teams were free to publish their own results with no embargo.

*Submission.* Teams registered on the ReXrank leaderboard and submitted predictions as a link to a single ZIP archive on Google Drive. The archive held one NIfTI file per test scan, with one binary mask channel per finding in the order given in the test file. Submissions were downloaded and scored automatically every two hours, and the leaderboard showed each submission's scores on the public half of the test set. There was no separate test phase and no limit on the number of submissions, but teams were asked not to submit again until their previous submission had been scored. Teams made between 1 and 242 submissions (median 4). The submission used for the final ranking is described in Section 2.4.

*Timeline.* The ReXGroundingCT training set and original validation scans had been public since 28 July 2025, and the 150 additional validation scans were released on 13 June 2026. Registration and submission opened on 14 June 2026, and the first submission was scored on 2 July 2026. By the deadline of 14 September 2026, 11:59 PM ET, 120 teams had registered. Results were announced at the MICCAI 2026 session on 27 September 2026.

*Ethics, data use, code, and conflicts of interest.* All scans come from CT-RATE, a public dataset of de-identified chest CT scans released under its own ethics approval (Hamamci et al., 2026b), so no additional approval was required for the challenge. The ReXGroundingCT data, including the annotations used in the challenge, are distributed under a CC BY-NC-SA 4.0 licence. The evaluation code that produced the rankings is public with the dataset (`https://huggingface.co/datasets/rajpurkarlab/ReXGroundingCT`). Teams were encouraged but not required to release their code, and the top teams were required to describe their methods in sufficient detail for this paper. The challenge had no sponsors or prizes. Access to the test-set reference masks was restricted to members of the Rajpurkar Lab at Harvard Medical School, who were not eligible to participate.

## Appendix B. Participating team method descriptions

Each of the four winning teams described its method using a common template whose headings follow the BIAS guideline. We edited the descriptions lightly for length and style, and report scores on the full test set.

*Appendix B.1. Team 1*

*Data.* Segmentation training used the 2,992 ReXGroundingCT training scans with their findings and masks. The image encoder, CIRCLE, was pretrained on 403,500 non-contrast chest CT scans from 288,740 patients, paired with radiology reports. Additional in-house data comprised 950 scans from the CIRCLE cohort at Zhongshan Hospital, Fudan University, Shanghai, China: 800 scans with pulmonary nodules and 150 challenging scans covering lung cancer, pneumonia and other infections, and less common thoracic conditions. Positive finding descriptions were extracted from each report, and annotation engineers with medical backgrounds exhaustively delineated every visible abnormality corresponding to each finding, yielding paired descriptions and 3D masks. No test-set images or text were used at any stage, including the pretraining of components.

*Pre-processing.* Volumes were resampled to $0.65 \times 0.65 \times 1.15$ mm, with an input size of $384 \times 320 \times 224$ voxels.

*Architecture and text encoder.* CIRCLE pretraining jointly learns image–report alignment and abnormality recognition. A 3D adaptation of EfficientNetV2-XL (Tan and Le, 2021) and a Chinese BERT-based report encoder project CT volumes and reports into a shared 512-dimensional space. A SigLIP-style contrastive objective (Zhai et al., 2023) aligns image–report pairs, while a parallel classification head predicts 37 thoracic abnormalities from report-derived labels with binary cross-entropy. For grounding, the pretrained image encoder was integrated into VoxTell (Rokuss et al., 2026) (Figure B.1): a prompt decoder conditions segmentation on features from a frozen Qwen3-Embedding-4B text encoder (Zhang et al., 2025b), and the image encoder and segmentation decoder are fine-tuned.

*Training.* Dice plus binary cross-entropy loss, AdamW with an initial learning rate of $3 \times 10^{-4}$, and a batch size of 24. Training took about two days on eight H20 GPUs.

*Ensembling.* Probability maps from checkpoints of different epochs were averaged voxel-wise.

*Post-processing.* The averaged probability map was thresholded at 0.5. No other post-processing was applied.

*Inference time and hardware.* About 20 s per scan per finding on one H20 GPU.

*Final submission and lessons.* The final submission is listed as "CIRCLE" on the leaderboard. On a 50-case validation subset, pretraining raised mean Dice from 0.277 with random initialization to 0.327, against 0.282 for VoxTell, and the in-house data raised it further to 0.356. Pretraining and additional annotations therefore helped, whereas the same architecture trained from scratch did not improve on VoxTell.

*Code and weights.* Code: `https://github.com/uii-open-source/CIRCLE/`. Model weights are not publicly released.

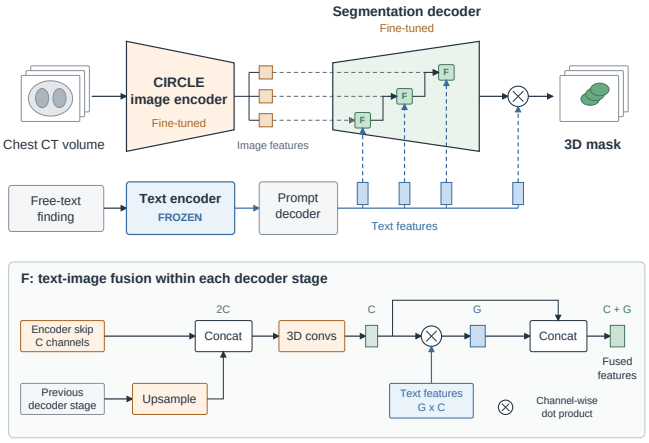



Figure B.1: Team 1: the pretrained CIRCLE image encoder combined with a VoxTell-based decoder. The image encoder and segmentation decoder are fine-tuned, and the text encoder is frozen. Inset F shows the text–image fusion within each decoder stage.

*Appendix B.2. Team 2*

*Data.* The challenge training set (2,992 scans with findings and masks), with the 200-scan validation set used only for model selection. Annotations were used unmodified. No external or in-house CT data, pseudo-labels or derived anatomy masks were used in the final system. The only externally pretrained component was Qwen3-Embedding-4B (Zhang et al., 2025b) (Apache 2.0), used frozen, and the image network was trained from scratch. Test-set scans and finding texts were not used for training or model selection. Whether any such material was part of the text encoder's own pretraining data is not known.

*Pre-processing.* Following nnU-Net preprocessing (Isensee et al., 2021), volumes were oriented to LPS and resampled to about $0.7 \times 0.7 \times 1.0$ mm. Intensities were clipped to $[-1000, 197]$ HU and normalized with a mean of $-549$ HU and a standard deviation of 313 HU. Training used random foreground-centred crops with zero padding, with patch sizes from $224 \times 224 \times 128$ to $448 \times 448 \times 256$ voxels.

*Architecture and text encoder.* The model, Grounder, uses a 3D nnU-Net residual-encoder (ResEnc-L) encoder (Isensee et al., 2024) and a five-stage decoder. The frozen Qwen3-Embedding-4B converts each finding into a 2,560-dimensional embedding, and two-layer MLPs generate scale and shift parameters that condition the decoder features through FiLM (Perez et al., 2018), $F' = \gamma F + \beta$.

*Training.* Dice plus cross-entropy loss, AdamW with a fixed learning rate of $10^{-4}$, and a batch size of one. Models generally converged within 20–30 epochs. Image augmentation included rotation, scaling and intensity transformations, but no flipping. In-distribution text augmentation replaced a finding with a real finding from another training case that matched it in location and finding type, keeping the original CT and mask. Each model trained on one H200 GPU for about 4–7 days, depending on patch size.

*Ensembling.* Thirteen independently trained models differing in patch size, selected on validation performance, with voxel-wise probabilities averaged equally.

*Post-processing.* Probabilities were thresholded at 0.5. No connected-component filtering, anatomical restriction, or rewriting of the text at inference was applied.

*Inference time and hardware.* About 5 s per scan per finding on one H200 GPU.

*Final submission and lessons.* The final submission was Grounder_1.4n (full test set: Dice 0.365, hit rate 0.763, instance F1 0.343). Larger anatomical context, in-distribution text augmentation and diverse patch sizes improved grounding. Adding lobe maps from TotalSegmentator (Wasserthal et al., 2023) or category embeddings did not help, and augmentation with negative text was unreliable because the training annotations are partial.

*Code and weights.* The repository `https://github.com/khuhm/Grounder` (MIT licence) will hold the code and trained weights, which have not yet been released.

### *Appendix B.3. Team 3*

*Data.* The challenge training (2,992 scans) and validation (200 scans) sets. External data were 8,000 sentence–mask pairs from CT-RATE (Hamamci et al., 2026b), with pseudo-masks of nodules and effusions from TotalSegmentator (Wasserthal et al., 2023), used by two ensemble members (one of them only the 1,540 effusion pairs). In-house data were used only for the pretraining of TAP-CT (Veenboer et al., 2025) (105,000 CT volumes). Pretrained weights were VoxTell v1.1 (Rokuss et al., 2026), Qwen3-Embedding (Zhang et al., 2025b), TAP-CT-B-3D and Qwen2.5-7B (Yang et al., 2025). Derived data were TotalSegmentator lobe masks, a lobe mapping of all finding sentences by Qwen3-32B (Yang et al., 2025), synonym-based text augmentation and Qwen2.5 paraphrases. Test-set exposure: the CT-RATE pool incidentally included test studies, so one member saw 80 of the 300 test studies and the other member 10 of these, as image, report sentence and pseudo-mask, but never the challenge masks or sentences. Retraining both members with these studies replaced by other CT-RATE scans changed the final masks equally on exposed and unexposed studies (Dice 0.976 vs. 0.974 between versions).

*Pre-processing.* VoxTell members use the native grid, z-scored intensities and random $192^3$ crops, one third of them centred on a finding. TAP-CT resamples to 1.5 mm isotropic voxels, clips to $[-1008, 822]$ HU, z-scores, and uses $24 \times 224 \times 224$ crops.

*Architecture and text encoder.* VoxTell: an nnU-Net residual encoder (Isensee et al., 2021, 2024) and a transformer decoder in which the frozen Qwen3-Embedding-4B sentence embedding queries the $12^3$ bottleneck tokens (440M trainable parameters). TAP-CT: a 3D ViT-B (87M parameters) conditioned on Qwen3-Embedding-0.6B through FiLM (Perez et al., 2018), followed by a convolutional decoder.

*Training.* VoxTell members were trained from VoxTell v1.1 for 250 epochs, five of them for up to 430, with deeply supervised Dice plus cross-entropy loss (Tversky loss in three members), SGD at $10^{-4}$ with polynomial decay, and a batch size of 8, taking about 7 h per 250 epochs on one H100. There was no image augmentation, and text augmentation replaced phrases with equivalent phrasings seen in the data ($p = 0.5$). The validation data were in every member's training set. TAP-CT members used Dice plus cross-entropy, AdamW and a batch size of 16, with 50,000 steps with the encoder frozen followed by 25,000 steps of fine-tuning (4.5 h per seed on one H100), five seeds, left–right flips, paraphrases and negatives with swapped locations.

*Ensembling.* The probability maps of twelve VoxTell members, which differ in initialization, loss, sampling and auxiliary data, are averaged and fused 0.8/0.2 with the mean of five TAP-CT seeds (Figure B.2).

*Post-processing.* The fused map is restricted to the named lobe (dilated by 16 voxels) or lung (12 voxels), thresholded per finding by a gradient-boosted regressor, and cleaned by a small 3D CNN that removes components predicted to lower Dice (both fitted on the validation set). Voxels claimed by several findings go to the most confident one. The only language model used at inference is the Qwen3-32B lobe mapping.

*Inference time and hardware.* On one H100, a VoxTell member takes 3.7 min per scan for all findings jointly, about 24 GPU-minutes per finding for the twelve members, and the five TAP-CT seeds together take 23 s per scan.

*Final submission and lessons.* The final submission was TAGS_TAPv3 (full test set Dice 0.362). Most helpful were TAP-CT as a second architecture, the dilated lobe restriction and the per-finding thresholds. Changes targeting location words, pseudo-labels and extra public data did not help. Four late post-processing additions each gained 0.001–0.004 on the validation set and the public half, yet the submission without them scored 0.365 on the full test set.

*Code and weights.* Code: `https://github.com/nki-radiology/TAGS-ReXGroundingCT` (Apache-2.0). Weights of all members: `https://huggingface.co/fomofo/TAGS-ReXGroundingCT` (CC BY-NC-SA 4.0, TAP-CT seeds CC BY-NC 4.0).

### *Appendix B.4. Team 4*

*Data.* The challenge training data and 200 validation scans, COVID-19-20 (180 scans) (Roth et al., 2022), OmniAbnorm-CT (52) (Zhao et al., 2026), a sparse interstitial lung disease dataset (91) (Depeursinge et al., 2012), and 18,652 CT-RATE scans (Hamamci et al., 2026b), excluding the test-set scans, with PatchChestCT annotations (Li et al., 2026) for 768 of them. The team reviewed the training annotations and edited 146 findings in 117 scans. Common errors were drawing artefacts, partly caused by the annotation tool, mismatches between

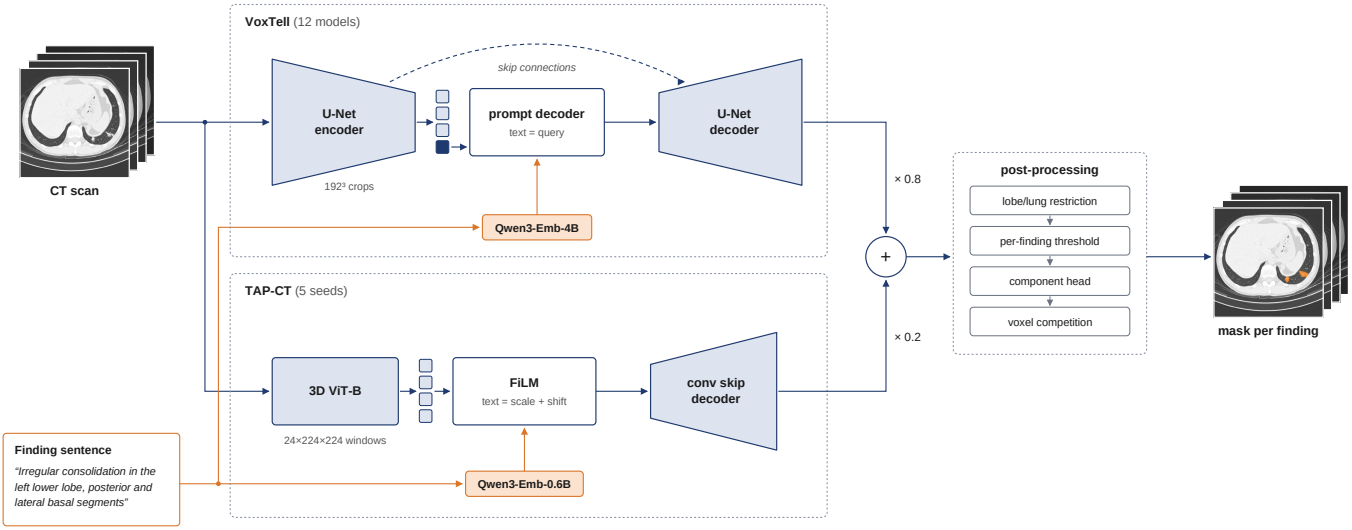



Figure B.2: Team 3: twelve VoxTell models and five TAP-CT seeds, each conditioned on the finding sentence, produce probability maps that are averaged per branch and fused 0.8/0.2, followed by four post-processing steps.

the location in the sentence and the mask, incomplete annotations such as missed nodules, arteries annotated as atelectasis or fibrosis, and the whole lung drawn as emphysema.

*Pre-processing.* Volumes were resampled to 0.75 × 0.75 × 1.25 mm and cropped to the lungs using TotalSegmentator lung masks (Wasserthal et al., 2023) with 15 mm margins. Intensities were clipped to $[-1024, 1000]$ HU and normalized as $(\mathrm{HU} + 232)/281$.

*Architecture and text encoder.* VoxTell v1.1 (Rokuss et al., 2026) with frozen Qwen3-Embedding-4B embeddings (Zhang et al., 2025b) and input crops of 288 × 288 × 192 voxels.

*Training.* Models were trained with an effective batch size of 8 across two RTX PRO 6000 GPUs, using AdamW with an initial learning rate of $10^{-5}$ and cosine decay unless stated otherwise, in three stages. First, two models were initialized from VoxTell v1.1 and trained on the challenge training set and complementary halves of the validation set, with the other half kept for checkpoint selection. The loss combined 0.3 binary cross-entropy, 0.7 Dice and 0.1 location KL divergence with deep supervision, and augmentation included intensity changes, noise, sharpening and paraphrased prompts. Second, training continued with the external segmentation datasets and with semi-supervised training on CT-RATE (Figure B.3). Positive findings were extracted from the CT-RATE reports with MedGemma (Sellergren et al., 2026), and two losses were added: a presence loss (weight 0.1), in which voxel logits are pooled with log-sum-exp and compared with whether the finding is present by binary cross-entropy, and a pseudo-mask loss from an EMA teacher (weight 0.2, decay 0.995). Third, each selected model was fine-tuned on the full validation set alone, with the supervised losses, for 500 steps at a constant learning rate of $3 \times 10^{-7}$.

*Ensembling.* The two models from the two validation folds were combined by averaging their probabilities.

*Post-processing.* For findings that name the left or right side, predictions confidently assigned to a lobe on the wrong side, according to TotalSegmentator, were removed.

*Inference time and hardware.* 5.90 s per finding (11.45 s per scan) on an RTX 3090, excluding input, output and post-processing.

*Final submission and lessons.* The final submission was the ensemble described above. Larger crops and fine-tuning on the curated data helped. Re-annotating non-focal categories such as bronchiectasis, and a Tversky loss to address the partial annotation of the training set, did not.

*Code and weights.* The edited data, code and weights will not be publicly released, but are available on request.

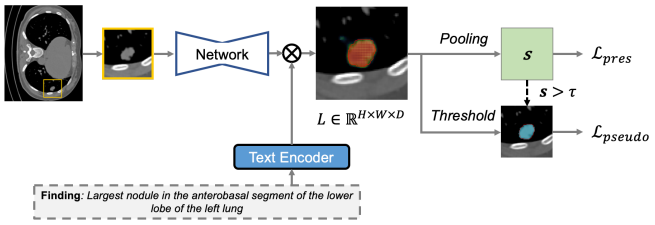



Figure B.3: Team 4: semi-supervised training on CT-RATE (stage 2). Voxel logits for the finding are pooled into a presence score trained against whether the finding is present, and thresholded predictions of an EMA teacher serve as pseudo-masks.

## Appendix C. Full leaderboard

Table C.1 lists all 48 ranked teams, and Figure C.1 shows how stable the ranking was under bootstrap resampling of the test scans.

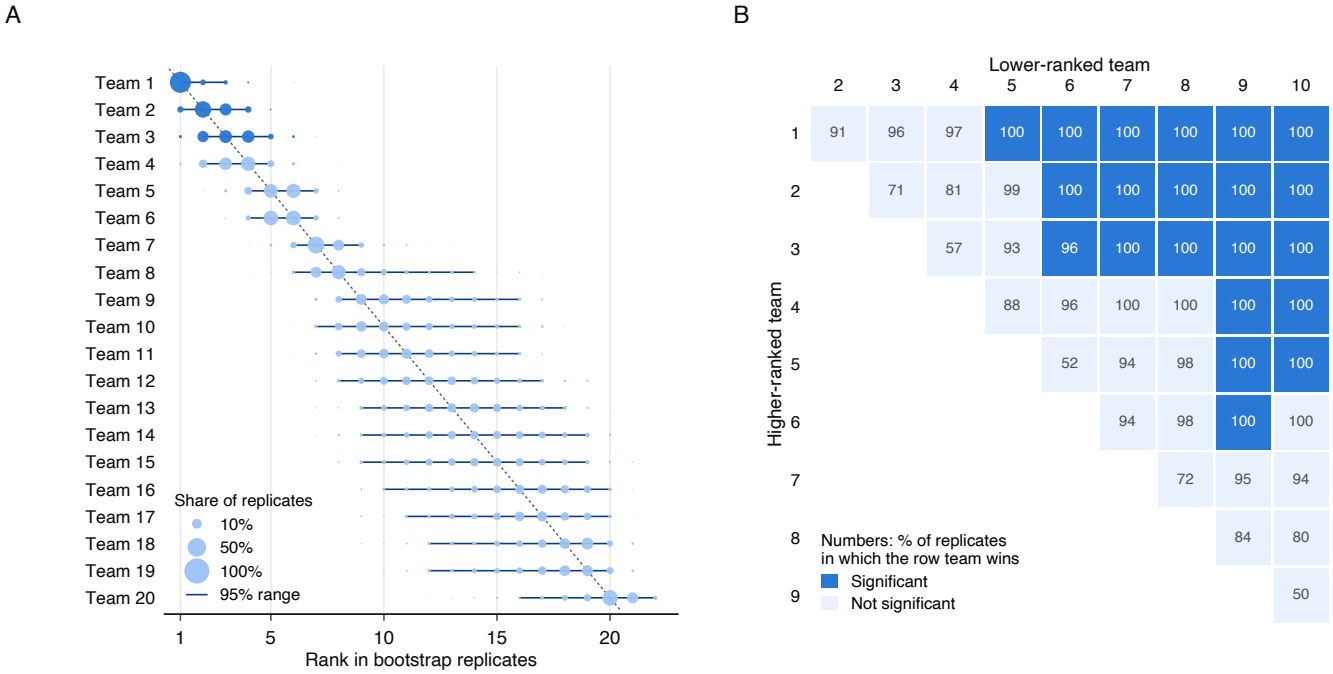



Figure C.1: Ranking stability, from 1,000 bootstrap replicates of the 300 test scans. A) Rank of the twenty best teams across replicates. Circle area is the share of replicates with that rank, the line spans the central 95% of ranks, and the dashed diagonal marks the observed rank; the three winning teams are shown in darker blue. B) Pairwise comparisons among the ten best teams, labelled by rank. Each number is the share of replicates in which the higher-ranked team scored higher. Filled cells are significant in a one-sided paired Wilcoxon test after Holm correction ($p < 0.05$). Confidence intervals of the mean Dice are given in Table 1.

## Appendix D. Examples from the controlled prompt experiment

## Appendix E. BIAS checklist

Table E.1 lists where this paper reports each item of the BIAS reporting guideline (Maier-Hein et al., 2020).

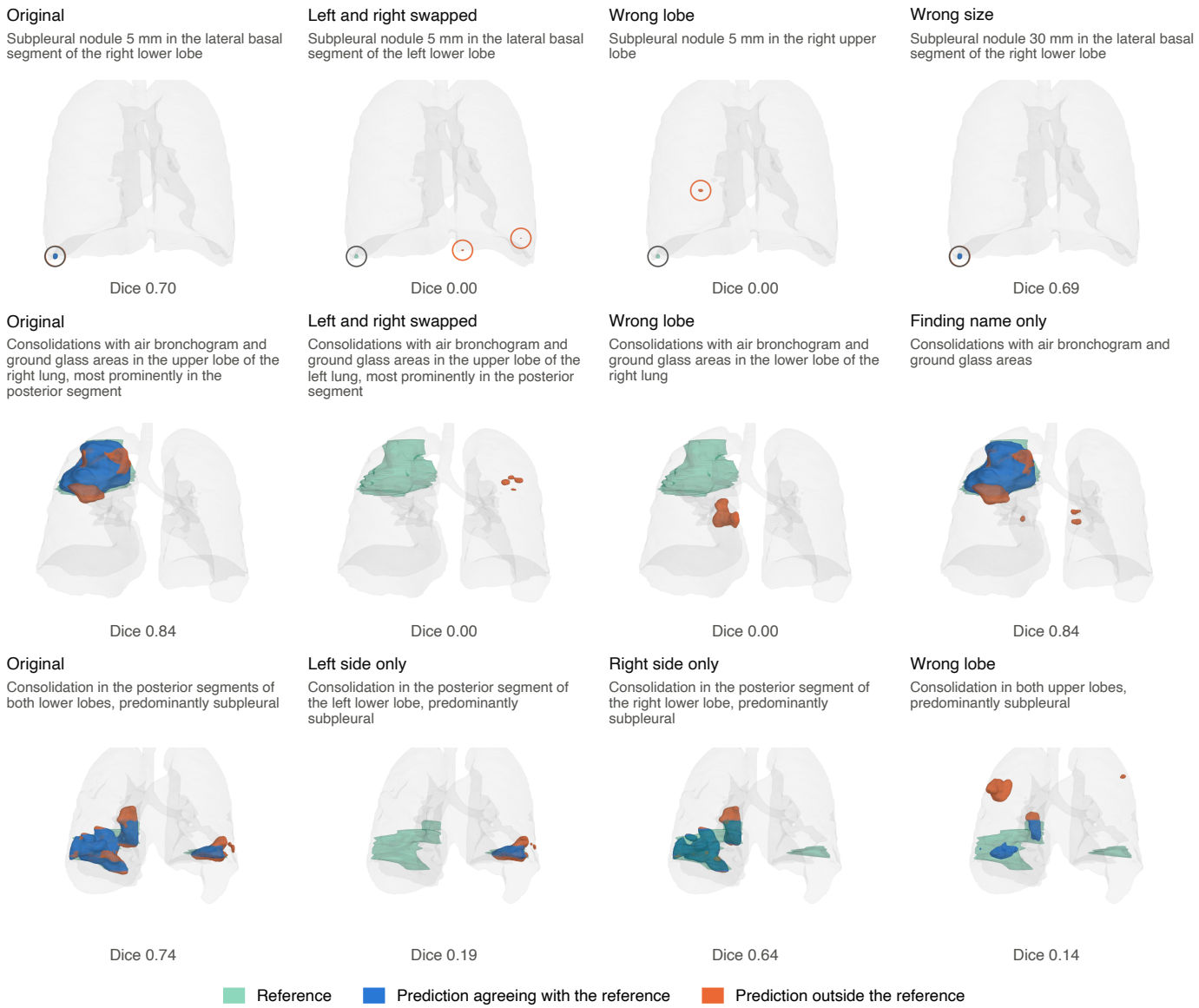



Figure D.1: Examples from the controlled prompt experiment (Team 2). Each row shows one test finding segmented with its original sentence (left) and with edited sentences. Masks are rendered in 3D inside the lungs and viewed from the front (patient's right on the left). The reference is shown in translucent green, and predicted surfaces are blue where they lie within 3 mm of the reference and orange elsewhere. Circles mark masks smaller than 1 mL. Dice is computed against the unchanged reference, so a prompt for one side of a bilateral finding (last row) can cover at most part of it.

Table C.1: Full leaderboard on the 300-scan test set, ordered by mean Dice. Teams are numbered by their final rank. CI, 95% bootstrap confidence interval of the mean Dice (1,000 replicates, Section 3.4). Submissions counts all submissions a team made to the leaderboard. Institutions are those of all team members and countries as given at registration, and – marks information that was not provided.

| Team | Institution | Country | Dice | Dice CI | Hit rate | Instance F1 | Submissions |
|---|---|---|---|---|---|---|---|
| Team 1 | United Imaging Intelligence, ShanghaiTech University, Zhongshan Hospital | China | 0.3755 | 0.352–0.399 | 0.7680 | 0.3410 | 6 |
| Team 2 | Gachon University, MEDAI | South Korea | 0.3654 | 0.344–0.387 | 0.7629 | 0.3428 | 19 |
| Team 3 | Netherlands Cancer Institute | Netherlands | 0.3621 | 0.342–0.383 | 0.7646 | 0.3336 | 38 |
| Team 4 | Yonsei University, NewcureM | South Korea | 0.3605 | 0.339–0.382 | 0.7766 | 0.3116 | 62 |
| Team 5 | Deepnoid | South Korea | 0.3526 | 0.330–0.375 | 0.7457 | 0.1899 | 21 |
| Team 6 | Siemens Healthineers, UCSF | USA | 0.3519 | 0.332–0.373 | 0.7646 | 0.2435 | 242 |
| Team 7 | Raidium | France | 0.3432 | 0.322–0.366 | 0.7526 | 0.2611 | 4 |
| Team 8 | German Cancer Research Center (DKFZ) | Germany | 0.3393 | 0.317–0.362 | 0.7423 | 0.2463 | 7 |
| Team 9 | Shanghai Jiao Tong University | China | 0.3329 | 0.313–0.354 | 0.7165 | 0.2481 | 12 |
| Team 10 | Hamad Bin Khalifa University | Qatar | 0.3329 | 0.311–0.355 | 0.6890 | 0.2985 | 44 |
| Team 11 | Friedrich-Alexander-Universität Erlangen-Nürnberg | Germany | 0.3317 | 0.311–0.353 | 0.7406 | 0.1971 | 112 |
| Team 12 | Microsoft Research India | India | 0.3292 | 0.308–0.352 | 0.7388 | 0.2421 | 5 |
| Team 13 | Shanghai Jiao Tong University, Shanghai Artificial Intelligence Laboratory, Medical Image Insights | China | 0.3272 | 0.307–0.348 | 0.7148 | 0.3256 | 16 |
| Team 14 | Duke University | USA | 0.3262 | 0.305–0.349 | 0.7371 | 0.2208 | 4 |
| Team 15 | Seoul National University | South Korea | 0.3258 | 0.306–0.346 | 0.7251 | 0.2835 | 19 |
| Team 16 | Medical University of Vienna | Austria | 0.3234 | 0.305–0.343 | 0.7491 | 0.2400 | 4 |
| Team 17 | Fudan University | China | 0.3231 | 0.302–0.343 | 0.7354 | 0.3402 | 42 |
| Team 18 | University of Campinas | Brazil | 0.3203 | 0.299–0.340 | 0.7131 | 0.1946 | 11 |
| Team 19 | LFAI | – | 0.3202 | 0.299–0.340 | 0.7165 | 0.1759 | 4 |
| Team 20 | Chongqing University, Hong Kong Polytechnic University, University of Massachusetts Lowell | China, USA | 0.3126 | 0.293–0.334 | 0.7079 | 0.2152 | 3 |
| Team 21 | University of Ulsan, National Institutes of Health, Asan Medical Center | South Korea | 0.3085 | 0.287–0.330 | 0.6821 | 0.2164 | 4 |
| Team 22 | Kosin University Gospel Hospital | South Korea | 0.2990 | 0.279–0.320 | 0.7010 | 0.1930 | 12 |
| Team 23 | Muroran Institute of Technology | Japan | 0.2868 | 0.266–0.310 | 0.6735 | 0.1746 | 3 |
| Team 24 | AMC | South Korea | 0.2855 | 0.264–0.307 | 0.6495 | 0.1799 | 1 |
| Team 25 | Microsoft Research | UK | 0.2748 | 0.256–0.295 | 0.6770 | 0.2241 | 4 |
| Team 26 | MBZUAI | UAE | 0.2737 | 0.252–0.297 | 0.6014 | 0.2141 | 6 |
| Team 27 | Korea University | South Korea | 0.2693 | 0.248–0.289 | 0.6615 | 0.1909 | 1 |
| Team 28 | Northwestern Polytechnical University | China | 0.2674 | 0.247–0.287 | 0.6460 | 0.1420 | 8 |
| Team 29 | – | – | 0.2673 | 0.247–0.288 | 0.6443 | 0.3114 | 3 |
| Team 30 | Robert Gordon University | UK | 0.2559 | 0.237–0.276 | 0.6478 | 0.1796 | 2 |
| Team 31 | University of Birmingham | UK | 0.2492 | 0.226–0.273 | 0.5447 | 0.1442 | 9 |
| Team 32 | University of Cambridge | UK | 0.2460 | 0.222–0.269 | 0.5292 | 0.1399 | 1 |
| Team 33 | Chinese University of Hong Kong | Hong Kong SAR | 0.2301 | 0.208–0.252 | 0.5172 | 0.1606 | 3 |
| Team 34 | Hangzhou Dianzi University, Tsinghua University | China | 0.2259 | 0.205–0.248 | 0.5619 | 0.1552 | 1 |
| Team 35 | University of Birmingham | UK | 0.2209 | 0.201–0.242 | 0.5069 | 0.1325 | 1 |
| Team 36 | Ant Group | China | 0.2201 | 0.200–0.243 | 0.5103 | 0.1520 | 1 |
| Team 37 | Independent | – | 0.2183 | 0.196–0.240 | 0.4708 | 0.1066 | 1 |
| Team 38 | Asan Medical Center | South Korea | 0.2170 | 0.196–0.240 | 0.5069 | 0.1092 | 2 |
| Team 39 | Pontificia Universidad Católica de Chile | Chile | 0.2136 | 0.192–0.235 | 0.5241 | 0.1500 | 8 |
| Team 40 | UCSF | USA | 0.2098 | 0.189–0.233 | 0.4845 | 0.1214 | 2 |
| Team 41 | Bangladesh University of Engineering and Technology | Bangladesh | 0.2096 | 0.189–0.233 | 0.4863 | 0.1218 | 2 |
| Team 42 | Independent | South Korea | 0.1943 | 0.177–0.214 | 0.5412 | 0.0982 | 2 |
| Team 43 | Erasmus University | Netherlands | 0.1936 | 0.174–0.214 | 0.4948 | 0.2051 | 1 |
| Team 44 | University of Wisconsin–Madison | USA | 0.1834 | 0.165–0.203 | 0.4897 | 0.1252 | 1 |
| Team 45 | Sichuan University of Science & Engineering | China | 0.1787 | 0.159–0.198 | 0.4003 | 0.0890 | 1 |
| Team 46 | Nagoya University | Japan | 0.1573 | 0.138–0.176 | 0.3763 | 0.0877 | 1 |
| Team 47 | Bangladesh University of Engineering and Technology | Bangladesh | 0.1545 | 0.139–0.171 | 0.4708 | 0.1727 | 2 |
| Team 48 | Technical University of Munich | Germany | 0.1461 | 0.130–0.162 | 0.4192 | 0.0542 | 2 |

Table E.1: Location of each BIAS checklist item in this paper.

| Item | Description | Location |
|---|---|---|
| 1 | Title conveys the essential information on the challenge | Title |
| 2 | Abstract: purpose, design and main results | Abstract |
| 3 | Keywords | Keywords |
| 4a | Biomedical motivation | Section 1 |
| 4b | Technical motivation | Section 1 |
| 4c | Purpose of the challenge and gap addressed | Section 1 |
| 5a–b | Challenge name and acronym | Appendix A, Name and organizers |
| 6 | Organizing team | Appendix A, Name and organizers |
| 7 | Life-cycle type | Appendix A, Life cycle and platform |
| 8a–c | Event, platform and website | Appendix A, Life cycle and platform |
| 9a | Allowed user interaction | Section 2.3 |
| 9b | Policy on training data and pretrained models | Section 2.3 |
| 9c | Participation of organizers | Appendix A, Participation, awards and publication |
| 9d | Award policy | Appendix A, Participation, awards and publication |
| 9e | Publication of results | Appendix A, Participation, awards and publication |
| 10a–b | Submission method and frequency | Appendix A, Submission; Section 2.4, Ranking |
| 11 | Timeline | Appendix A, Timeline |
| 12 | Ethics approval | Appendix A, Ethics, data use, code |
| 13 | Data usage agreement | Appendix A, Ethics, data use, code |
| 14a–b | Code availability (organizers and participants) | Appendix A; Declarations; Appendix B |
| 15 | Conflicts of interest | Declarations |
| 16 | Author contributions | Declarations |
| 17 | Field of application | Section 2.1, Task |
| 18 | Task category | Section 2.1, Task |
| 19a–b | Target cohort and challenge cohort | Section 2.2, Cohort |
| 20 | Imaging modality | Section 2.2, Cohort |
| 21 | Context information | Section 2.1, Task |
| 22a–b | Data origin and algorithm target | Section 2.2, Cohort |
| 23 | Property to be optimized | Section 2.1, Assessment aim |
| 24a–d | Acquisition devices, protocols, centres and operators | Section 2.2, Data source and Cohort |
| 25a | Definition of a case | Section 2.2, Data splits |
| 25b | Number of training, validation and test cases | Section 2.2, Data splits |
| 25c | Justification of the split sizes | Section 2.2, Data splits |
| 25d | Characteristics of the splits | Section 2.2; Figure 2 |
| 26a–d | Annotation method, instructions, annotators and merging | Section 2.2, Annotation |
| 27 | Pre-processing | Section 2.2, Pre-processing |
| 28a–b | Sources of error and inter-rater variability | Section 2.2, Sources of error |
| 29a–b | Metrics and their justification | Section 2.4, Metrics |
| 30a–c | Ranking method, missing results and justification | Section 2.4, Ranking |
| 31a–b | Statistical methods and justification | Section 2.4, Statistical analysis |
| 32 | Further analyses (methods) | Section 3.7 |
| 33a–b | Participating teams and method descriptions | Section 3.1, Section 3.2; Appendix B |
| 34 | Metric values | Table 1; Table C.1 |
| 35a | Rankings and number of test cases | Section 3.3; Table 1 |
| 35b | Results of the statistical analysis | Section 3.4; Figure C.1 |
| 36 | Results of further analyses | Section 3.5, Section 3.6, Section 3.7 |
| 37 | Summary of the results | Section 4 |
| 38a–b | Biomedical and technical impact | Section 4 |
| 39a | Is the task solved? | Section 4 |
| 39b | Analysis of individual cases | Section 3.6; Section 4 |
| 39c | Strengths and weaknesses of the methods | Section 3.2; Appendix B |
| 40 | Limitations | Section 4 |
| 41 | Future work | Section 4 |